UiA Universitetet i Agder

MAS508-G
Lab 1 – Introduction to MATLAB/Simulink

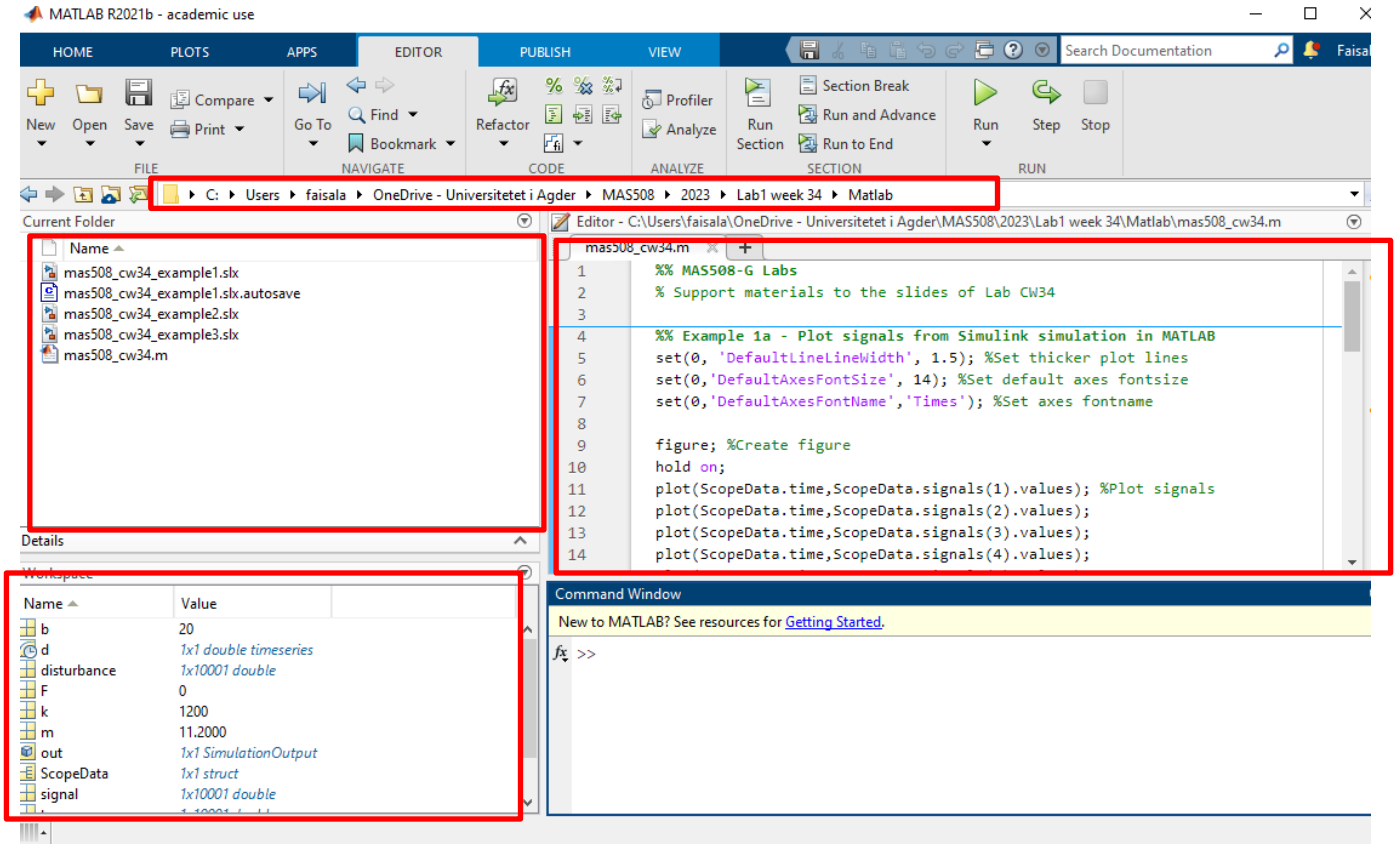


Lab -1 Introduction to MATLAB/Simulink

- **Introduction to MATLAB/Simulink**
- **Creating a model in Simulink**
  - **Simulink blocks**
  - **Solver configuration**
- **Examples and exercises**

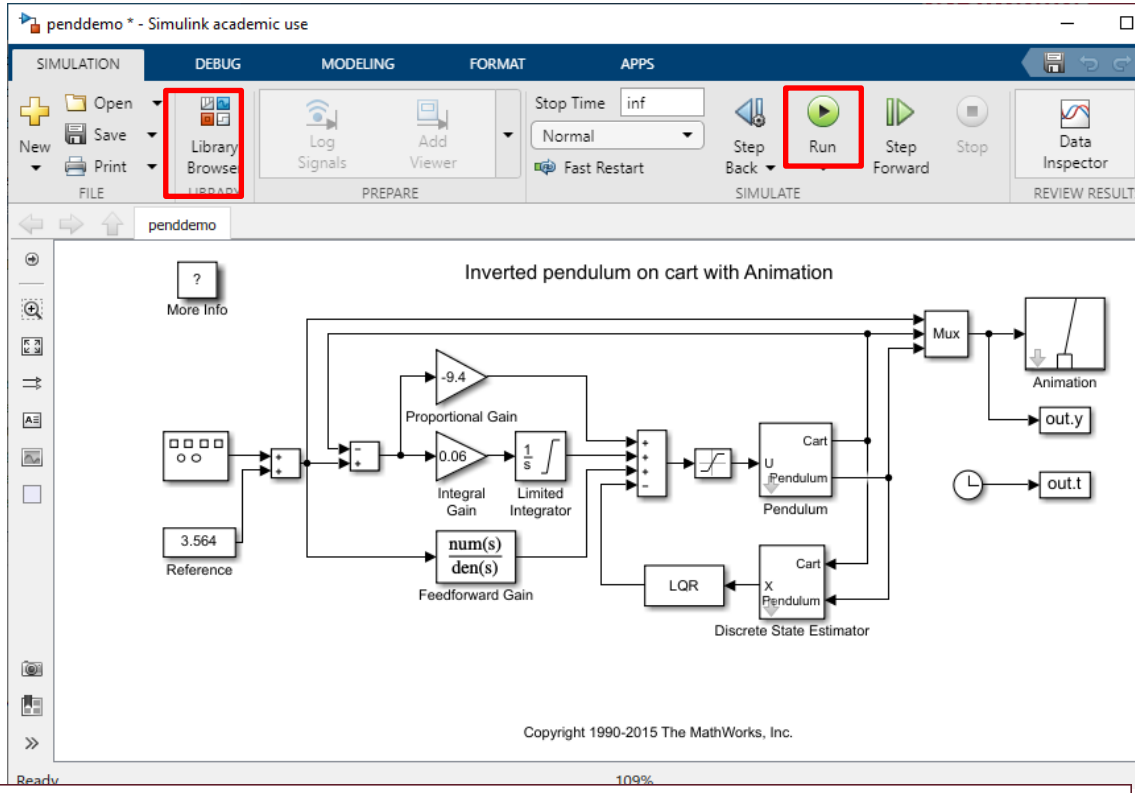
MATLAB Interface

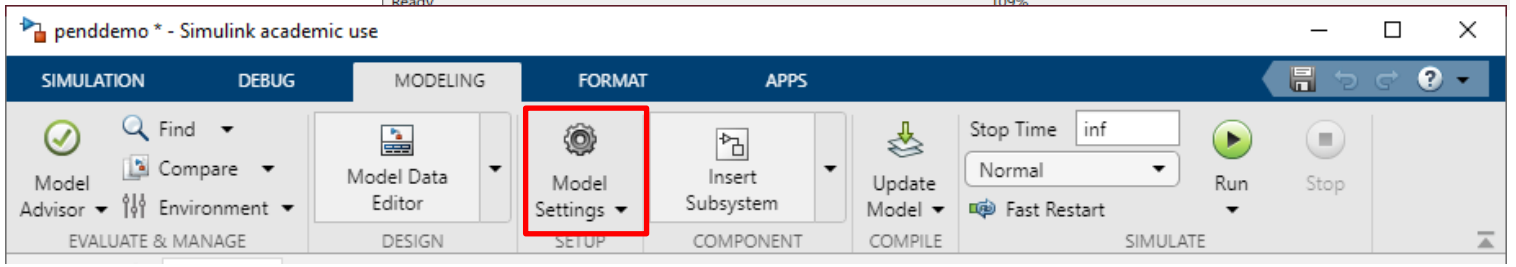
- **Code-based application**
- **Current Path/Folder**
- **Editor- Code Script Creation**
- **Command Window- Direct Code Execution**
- **Workspace - Variables**



Simulink Interface

- Click  or type Simulink in MATLAB command window
- GUI based interface
- Shared workspace with MATLAB
- Easy to set up simulations
- Drag & drop predefined blocks
- Avoid coding errors
- Display signal variables
- Real-Time co-simulation



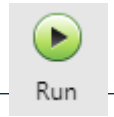


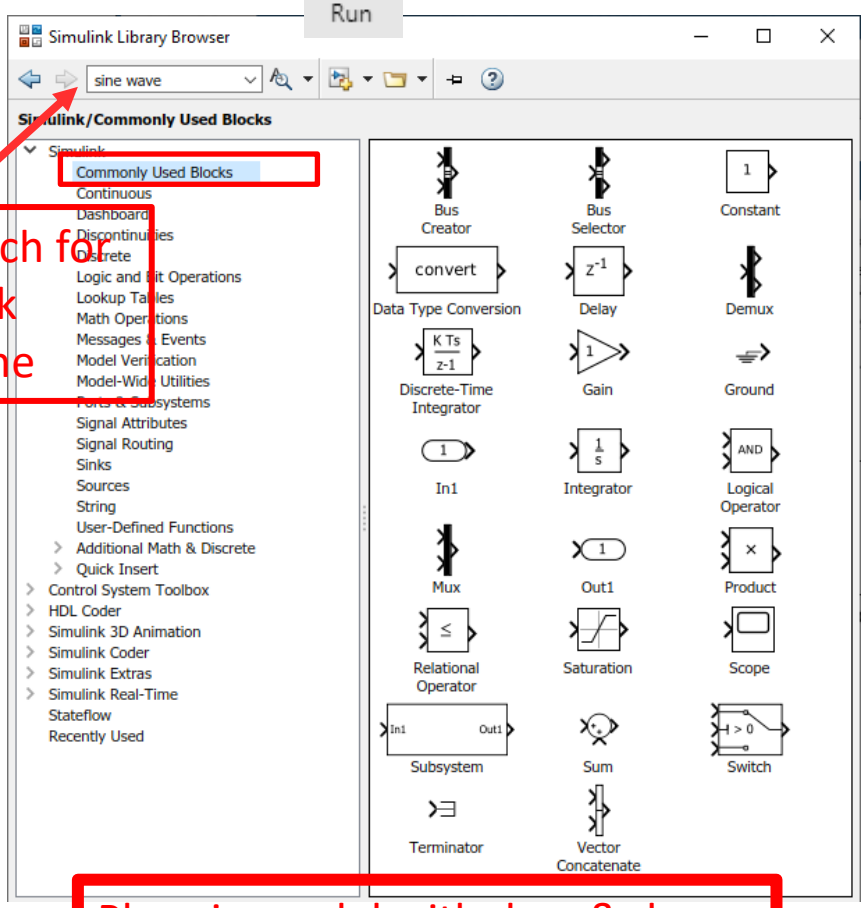
Creating Model in Simulink

- Create Simulink Model using Blocks from Library Browser
- Set Model Properties / Solver
- Run Simulation







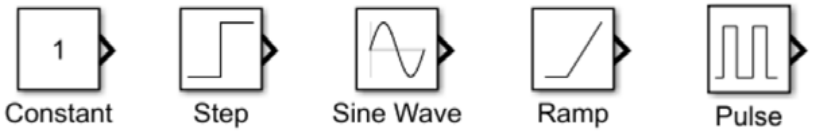


Search for block name

Place in model with drag & drop

Commonly used Simulink Blocks

Signals



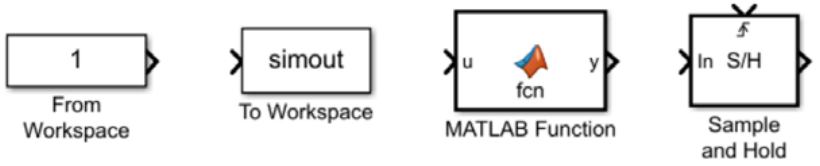
Mathematical Operators



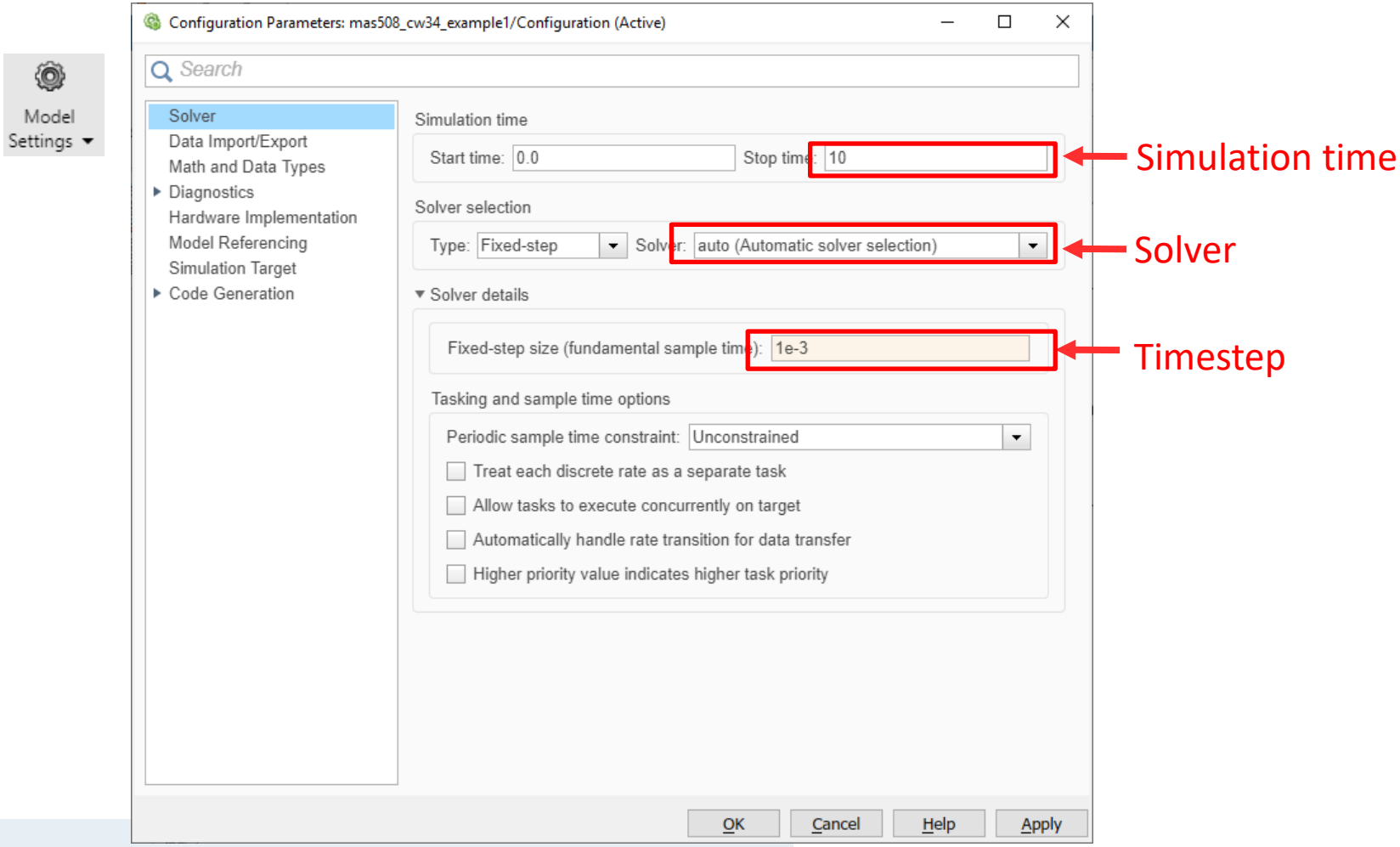
Signal modification and display



Other useful blocks



Solver Configuration

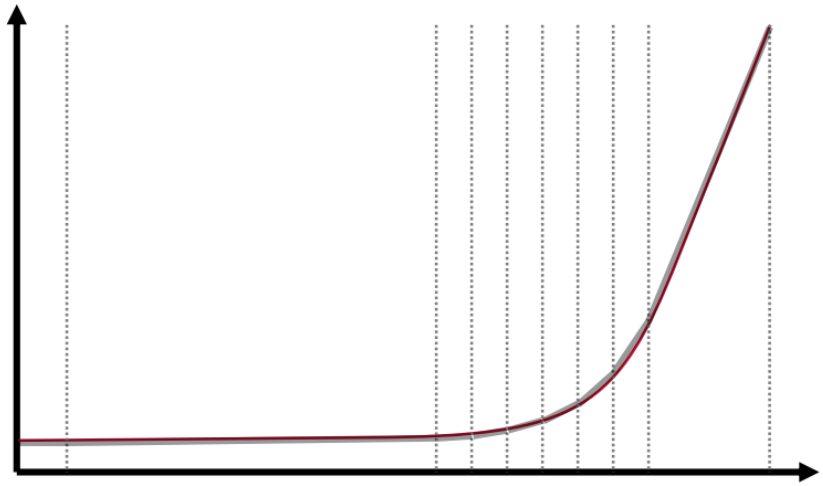


NOTE

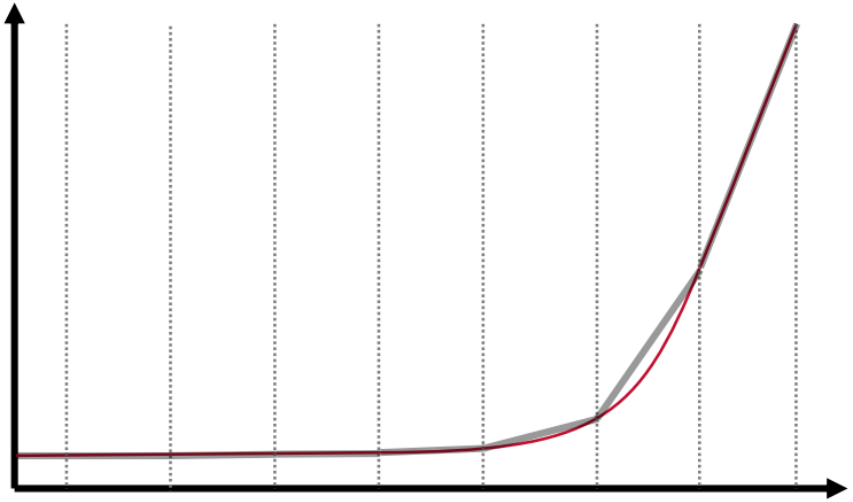
- Smaller step size → Longer simulation
- Bigger step size → Faster dynamics less captured

Solvers differences

- **Variable-step**
  - Ode45, ode23,ode 113 etc
  - Step-size→automatic according to model



- **Fixed-step**
  - ode5 (Dormand-Prince), ode4 (Runge-Kutta), ode1(Euler), etc.
  - Step-size→fixed defined by user



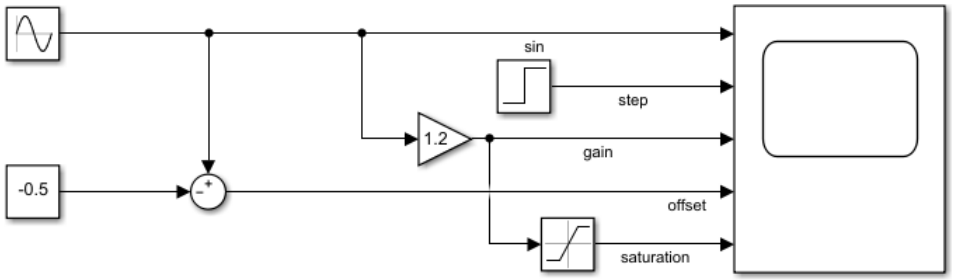
How to select an appropriate timestep

- Depends on the problem or system
  - Slow dynamics → higher step size. Example: heat transfer → seconds
  - Faster dynamics → smaller step size. Example: electric systems → ms/µs

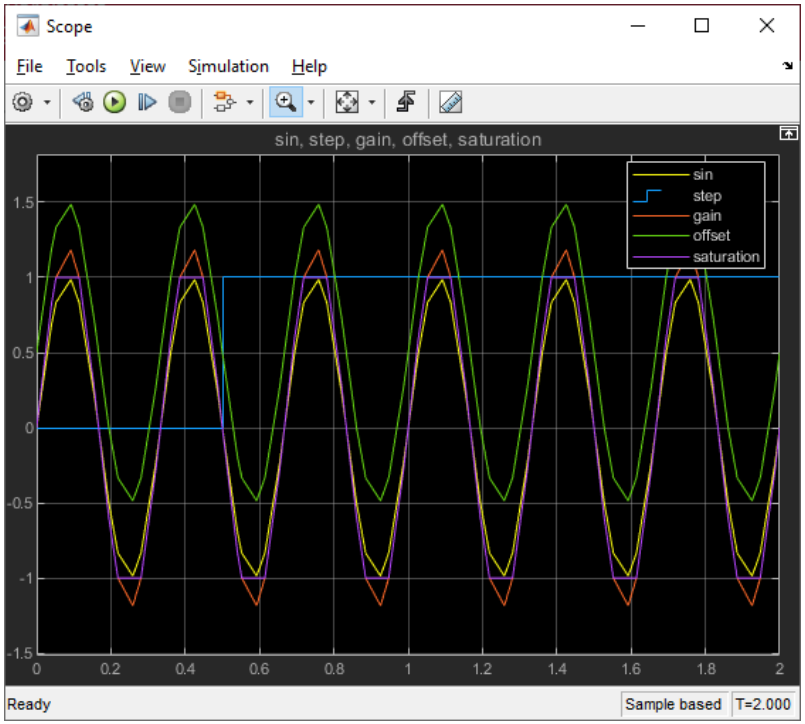
Example 1- Create and display signals

- Create a Simulink model to display the following signals:
  1) Sine wave with amplitude of 1, frequency of 3 Hz and no phase shift
  2) Step signal with initial/final value of 0/1 and step time of 0.5 s
  3) Signal 1) with an amplification (gain) of 1.2
  4) Signal 1) with an offset of -0.5
  5) Signal 3) limited between -1 to 1 (saturation)
- Show all signals in the same scope
- Name signals and display Legend in Scope
- Simulation time: 2s
- Evaluate the results for:
  - Variable-step with auto settings
  - Fixed-step with timestep 1e-2
  - Fixed-step with timestep 1e-3

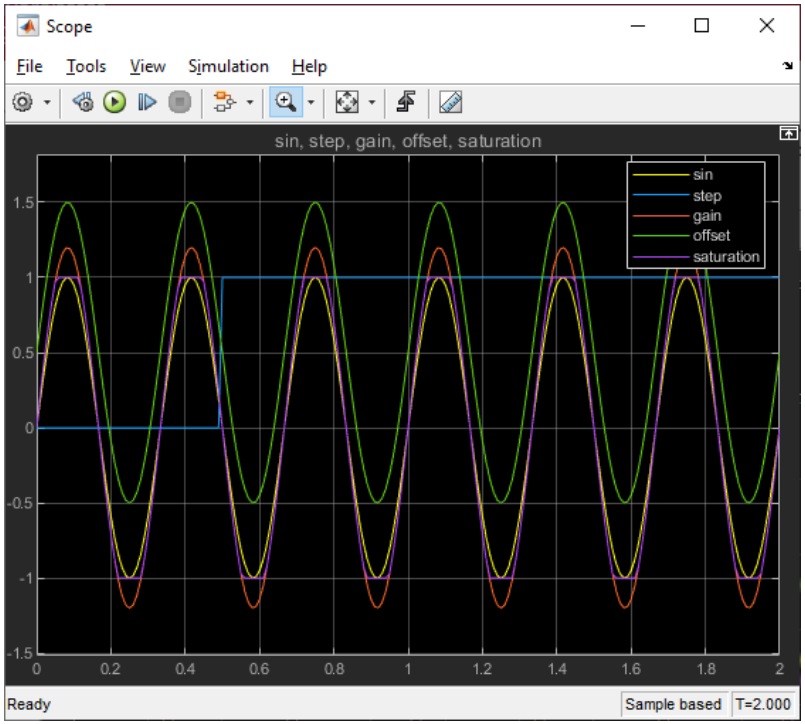
Example 1- Solution



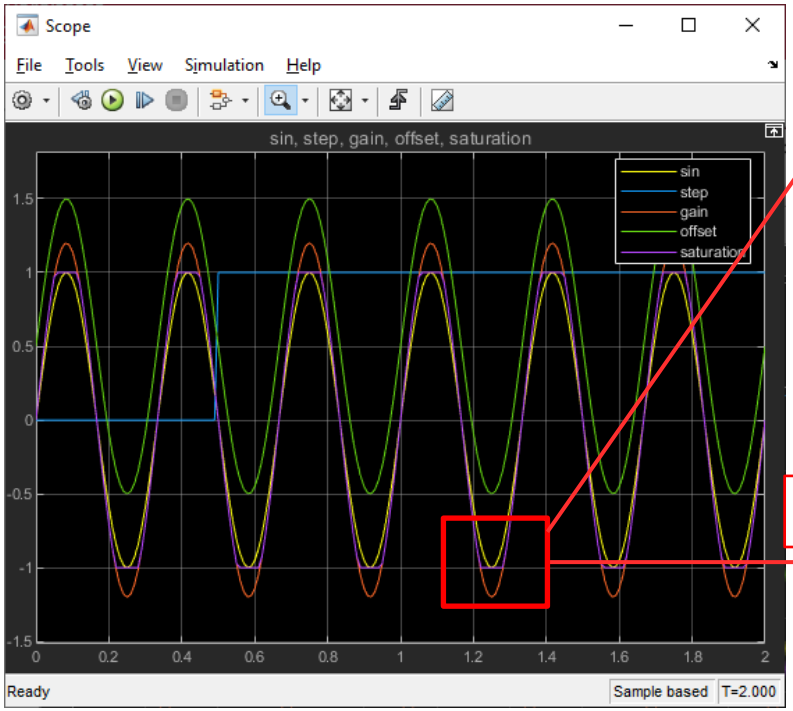
Variable-step with auto settings



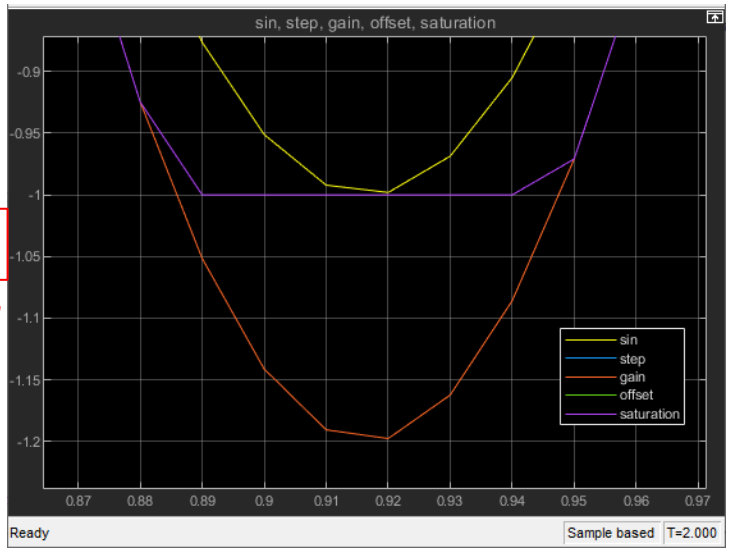
Fixed- step with timestep(Ts) =1e-2



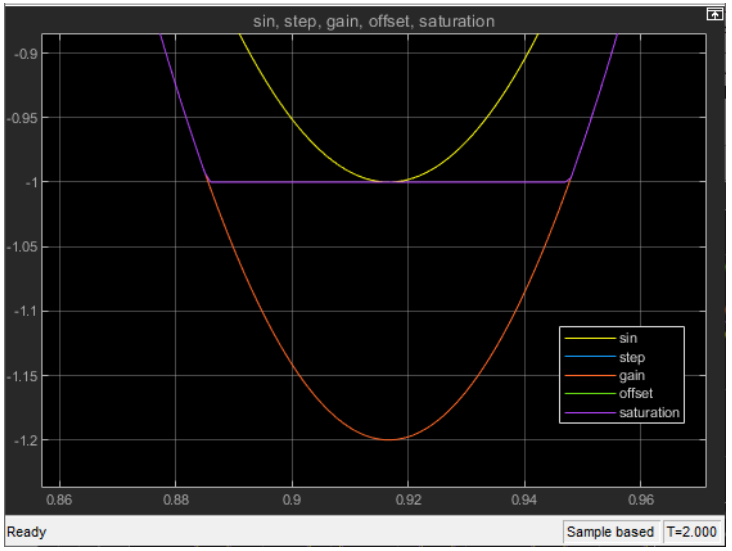
Timestep comparison



Ts=1e-2

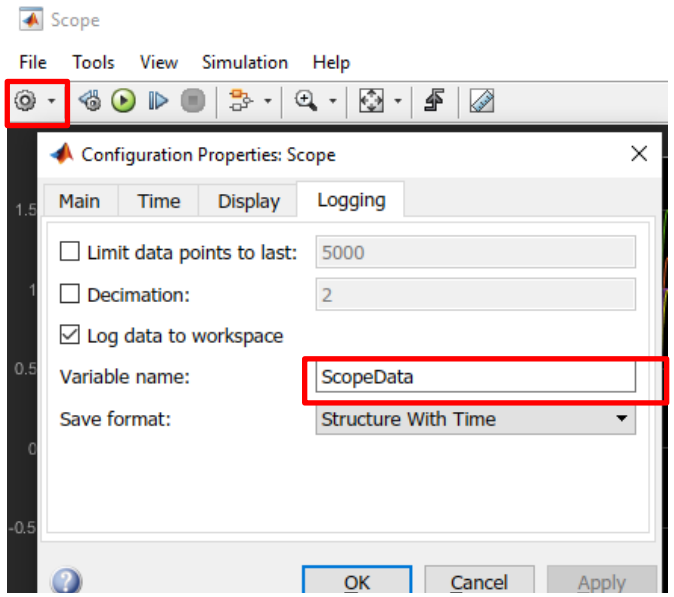


Ts=1e-3



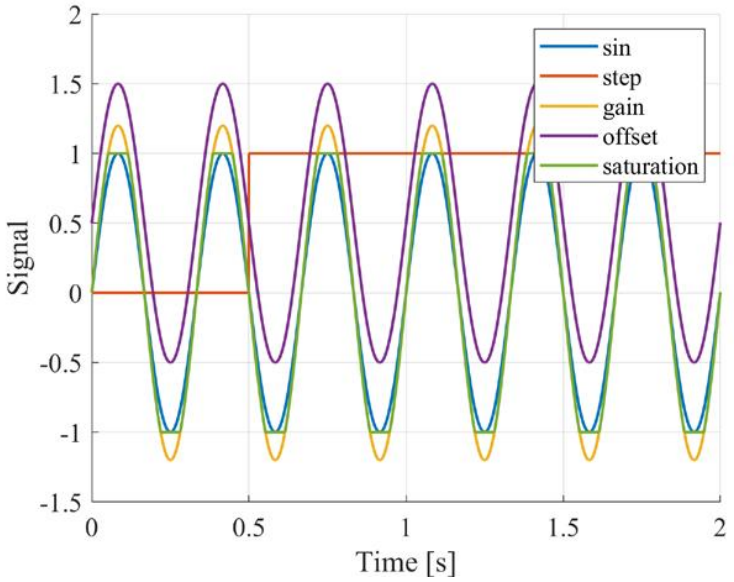
Plot ScopeData in MATLAB

- Untick Model Settings > Data Import/Export >Single simulation output
- Configure the Scope Configuration Properties > Logging and select Variable name



- Plot in MATLAB (example code)

```
set(0, 'DefaultLineLineWidth', 2); %Set thicker plot lines
figure; %Create figure
hold on;
plot(ScopeData.time,ScopeData.signals(1).values); %Plot signals
plot(ScopeData.time,ScopeData.signals(2).values);
plot(ScopeData.time,ScopeData.signals(3).values);
plot(ScopeData.time,ScopeData.signals(4).values);
plot(ScopeData.time,ScopeData.signals(5).values);
legend(ScopeData.signals.label); %Add legend
xlim([0 2]); %Set window limits on x-axis
ylim([-1.5 2]); %Set window limits on y-axis
grid on; %Turn on grid lines
```

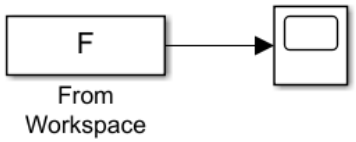


Use signals from MATLAB workspace in Simulink

- Create timeseries of the signal:

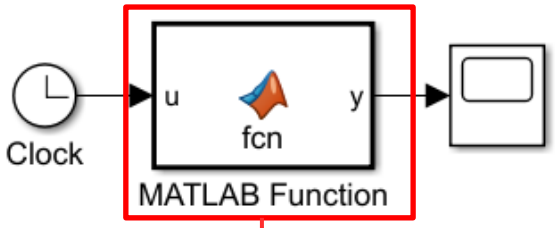
```
time=0:1e-3:2;
signal=2*sin(2*pi*t);
F=timeseries(signal,time);
```

- Use From Workspace block



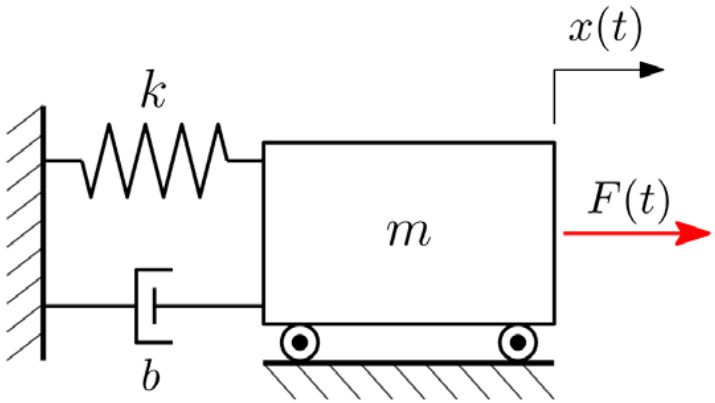
Use Matlab CODE IN Simulink

- Use MATLAB Function block




```
function y = fcn(u)
y = 2*sin(2*pi*3*u);
```

Example 2- Mass-spring-damper system



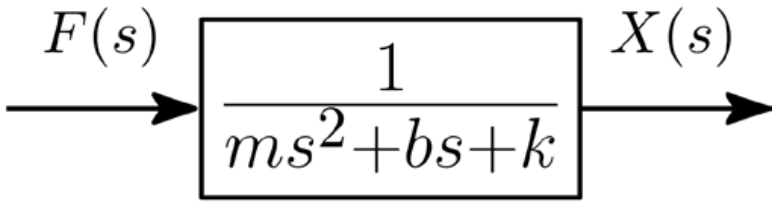
- **2nd order linear differential equation**

$$\sum F_x = m\ddot{x}$$

$$m\ddot{x} + b\dot{x} + kx = F$$

- **Transfer function**

$$G(s) = \frac{X(s)}{F(s)}$$



Example 2- Mass-spring-damper system

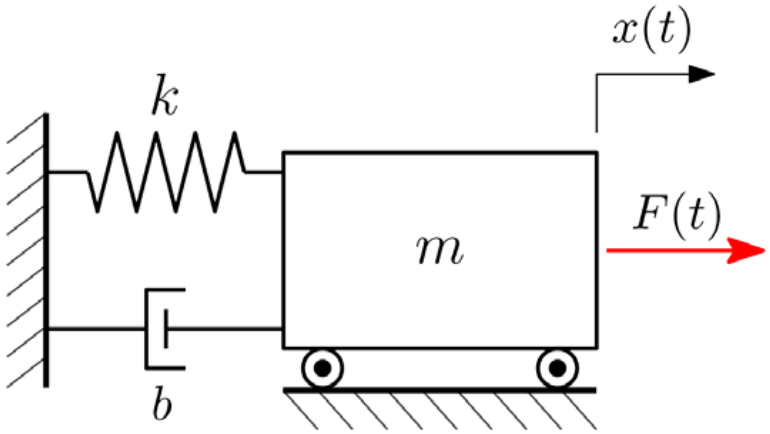
- **Create a Simulink model for a mass-spring-damper system**
- **Plot/display velocity and position with no external force**
  - **Initial conditions**

$$\dot{x} = 0 \text{ m/s}$$
$$x = 1.5 \text{ m}$$

  - **Initial conditions**

$$\dot{x} = 2 \text{ m/s}$$
$$x = 0 \text{ m}$$

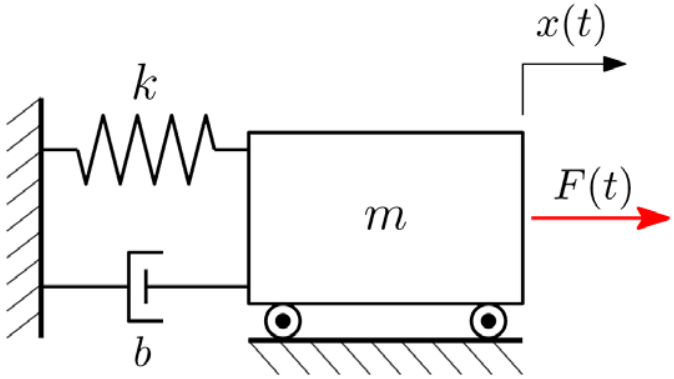
  - **Simulation time: 6 seconds**



System Properties

$$m = 11.2 \text{ kg}$$
$$b = 20 \text{ Ns/m}$$
$$k = 1200 \text{ N/m}$$

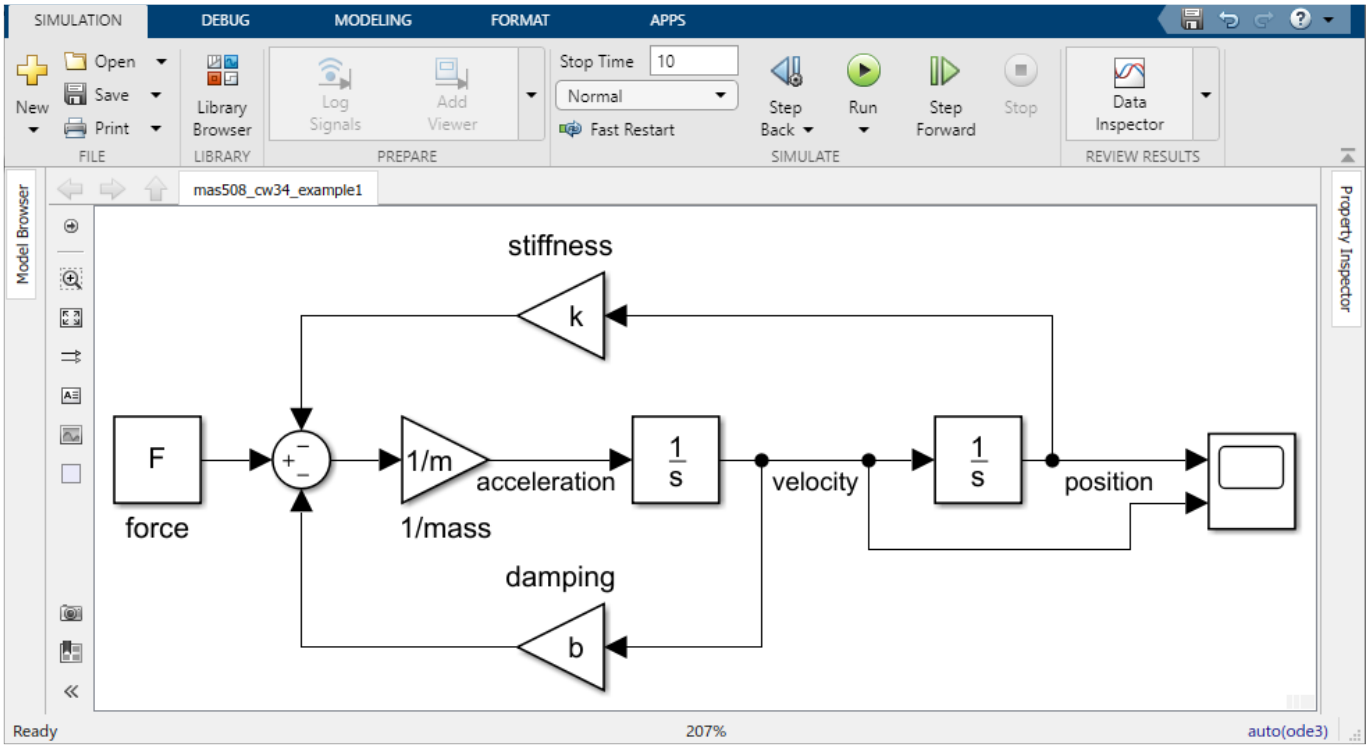
Example 2- "Forced motion"



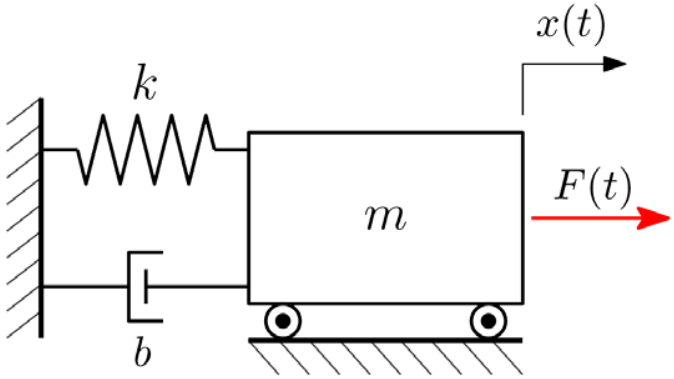
- **2nd order linear differential equation**

$$m\ddot{x} + b\dot{x} + kx = F$$

$$\ddot{x} = \frac{1}{m}(F - b\dot{x} - kx)$$



Example 2- "Free motion"

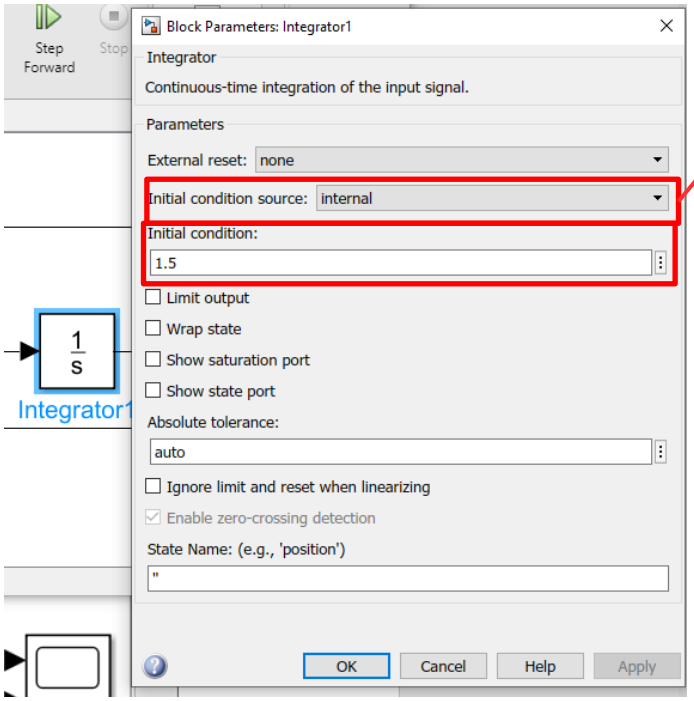


- **2nd order linear differential equation**

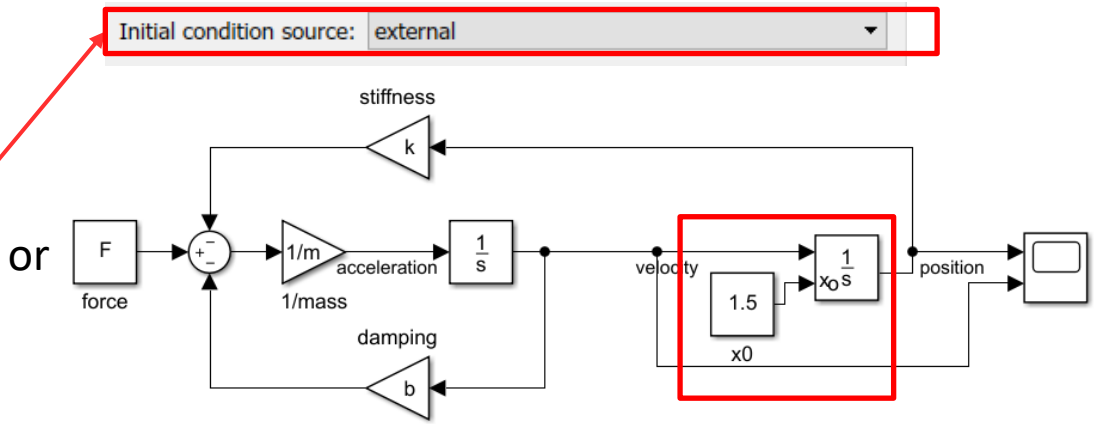
$$m\ddot{x} + b\dot{x} + kx = \underbrace{F}_{=0}$$

$$\ddot{x} = \frac{1}{m}(\underbrace{F}_{=0} - b\dot{x} - kx)$$

- **Initial conditions**



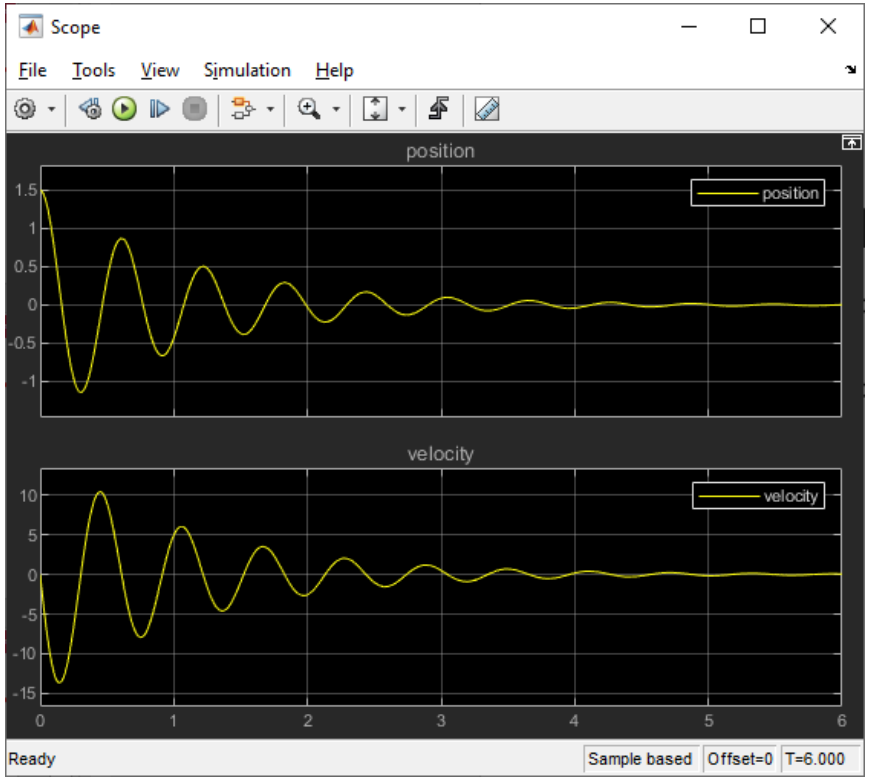
or



Example 2 - Solution

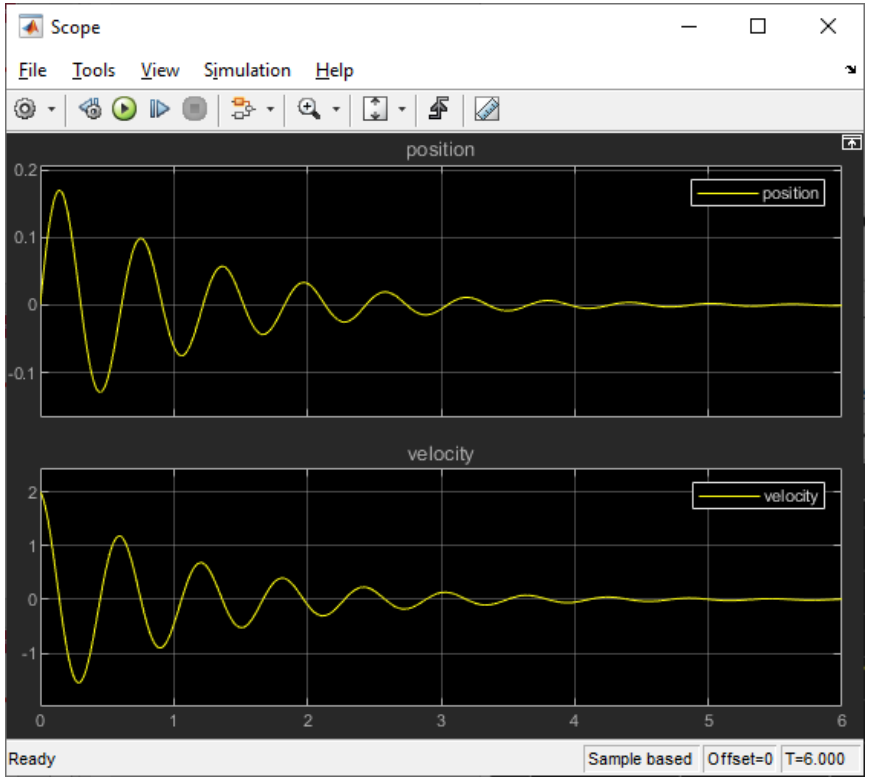
- **First initial conditions**

$$\dot{x} = 0 \text{ m/s}$$
$$x = 1.5 \text{ m}$$



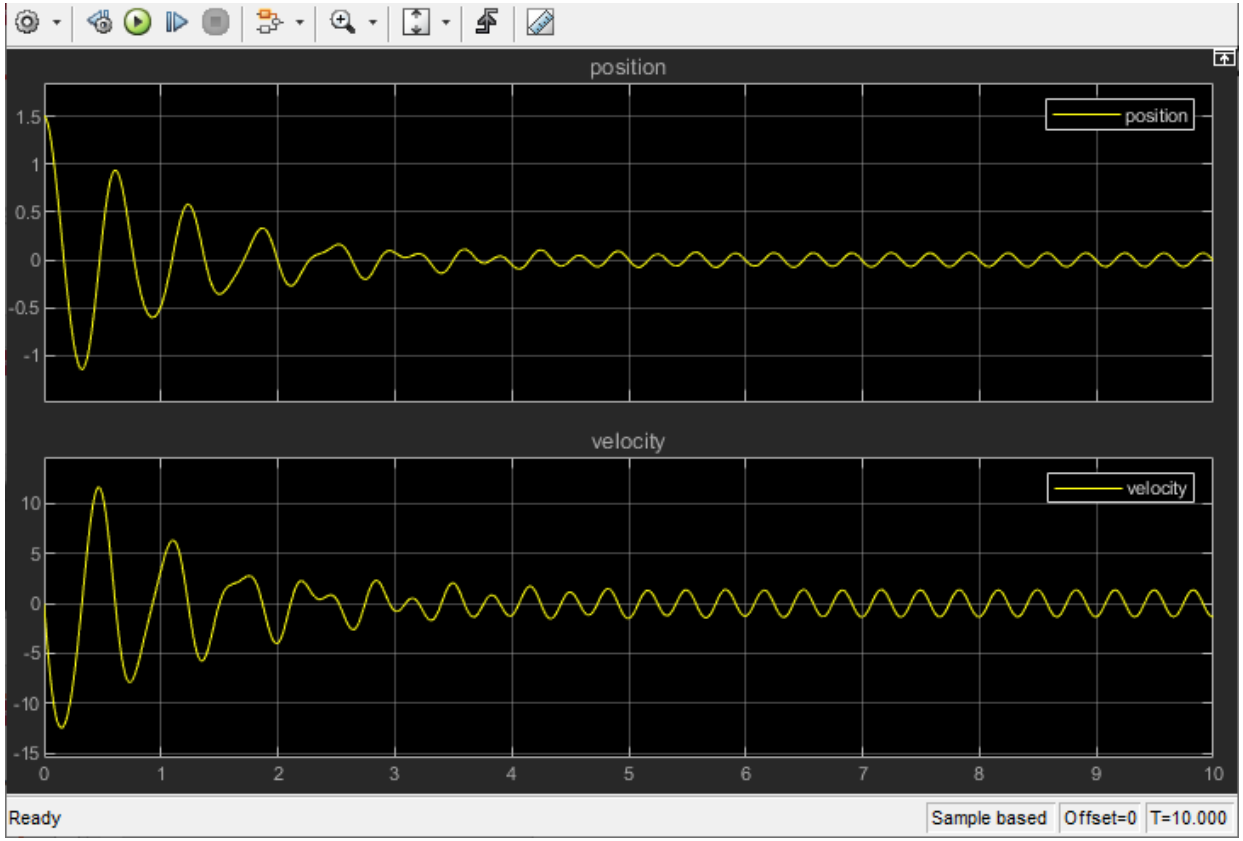
- **Second initial conditions**

$$\dot{x} = 2 \text{ m/s}$$
$$x = 0 \text{ m}$$

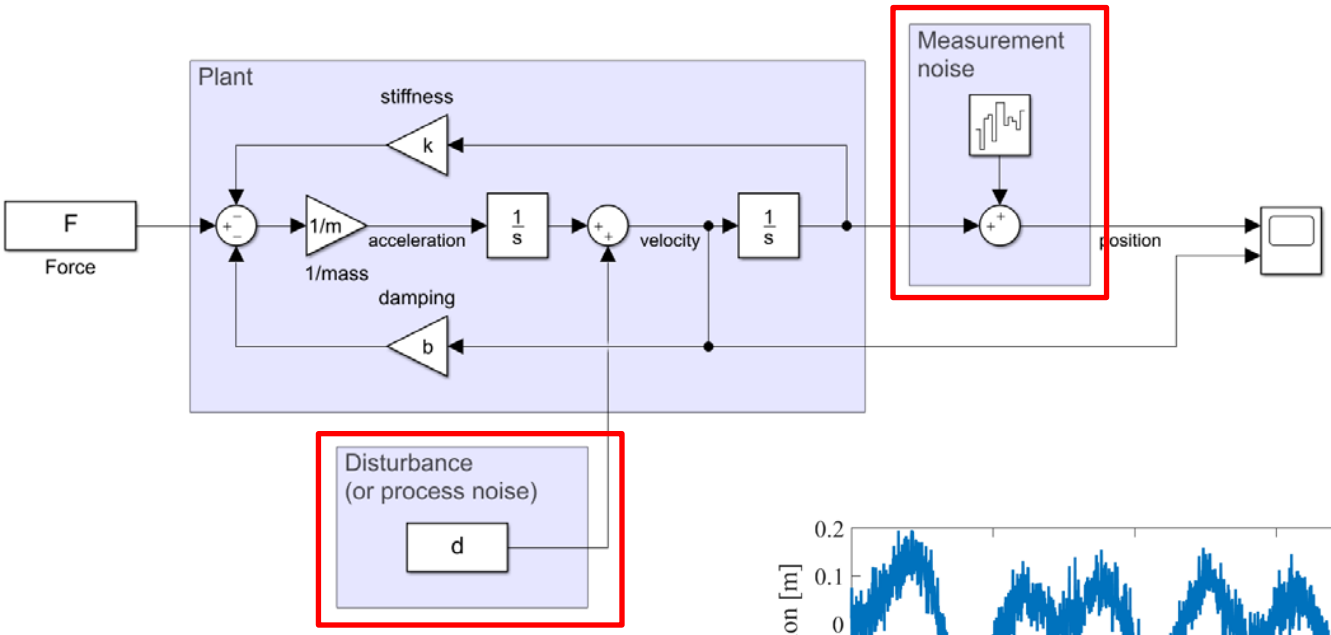


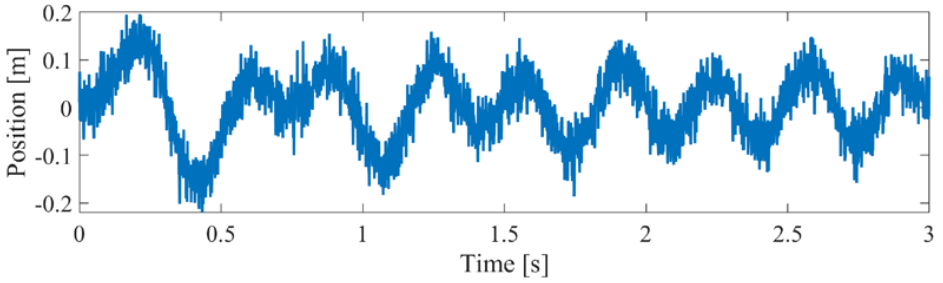
Example 2

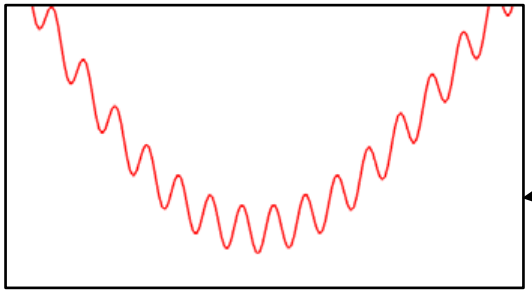
- **Add an external sinusoidal force and display position and velocity**
  - **Amplitude: 200N**
  - **Frequency: 3 Hz**
  - **Use first initial conditions**

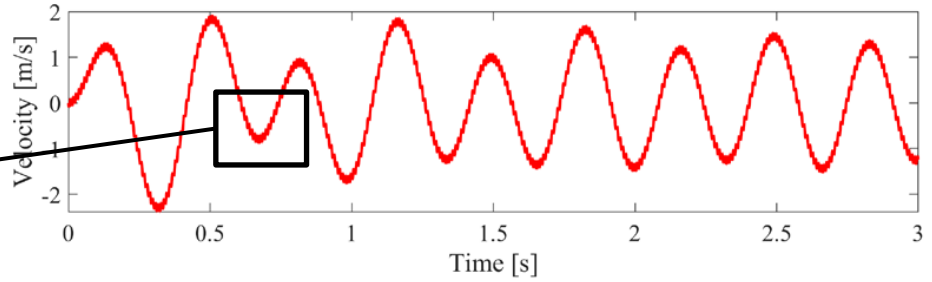


Example 2 – Process and measurement noise

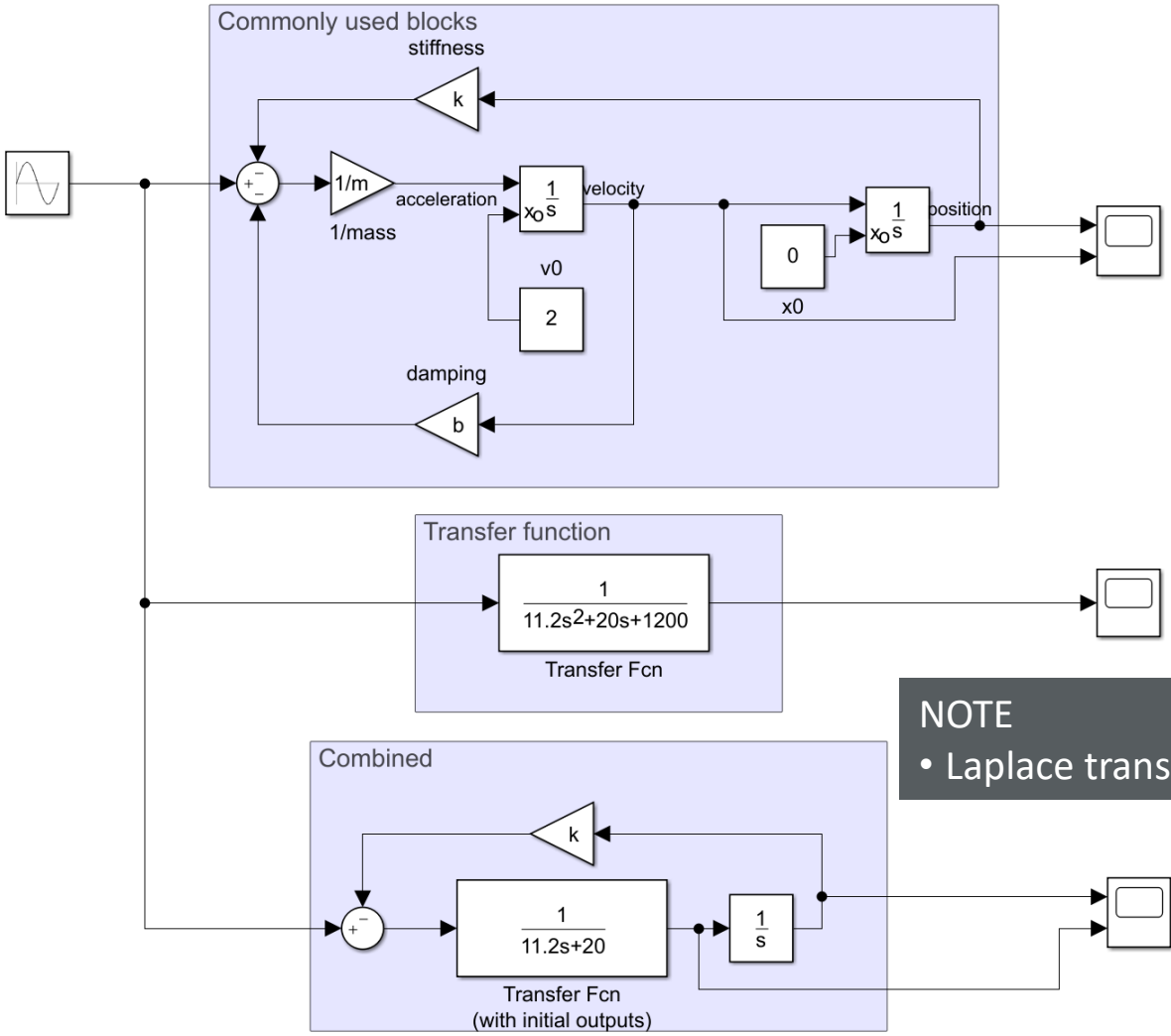








Example 3 – Other solutions for implementation



NOTE
- Laplace transform → Zero initial conditions